INTERNET OF THINGS

Module II — Communication Technologies & Protocols

Topics

- **Wireless & Wired Communication Technologies (Physical/Data-link layer):**

  ZigBee, Wi-Fi, 6LoWPAN, LoRa & NB-IoT

- **IoT Application-Layer Protocols:**

  MQTT, WebSocket, HTTP/HTTPS, AMQP, CoAP, XMPP

- **Identification Protocols**:

  IPv6 & URI

- **Network Architecture**:

  WSN and 4G/5G

1. Communication Technologies in IoT — The Big Picture

Every IoT device needs to talk to something — a nearby sensor, a gateway, or the internet. This 'talking' happens at the physical and data-link layer, which decides how raw bits actually travel between two nearby devices (like radio waves, cables, or infrared). IoT networks use two broad families of technology: wireless (no cables — Wi-Fi, ZigBee, LoRa, etc.) and wired (cables — Ethernet, RS-485, etc.). Most modern IoT deployments prefer wireless because sensors are cheap, battery-powered, and scattered across large areas where wiring is impractical.

Key Points

- Physical + Data-link layer = how bits move on the local network (LAN/PAN level), before the data even reaches the internet.
- A local IoT/M2M (machine-to-machine) network usually connects many small devices to one gateway.
- Choice of technology depends on range, power budget, data rate, and cost.

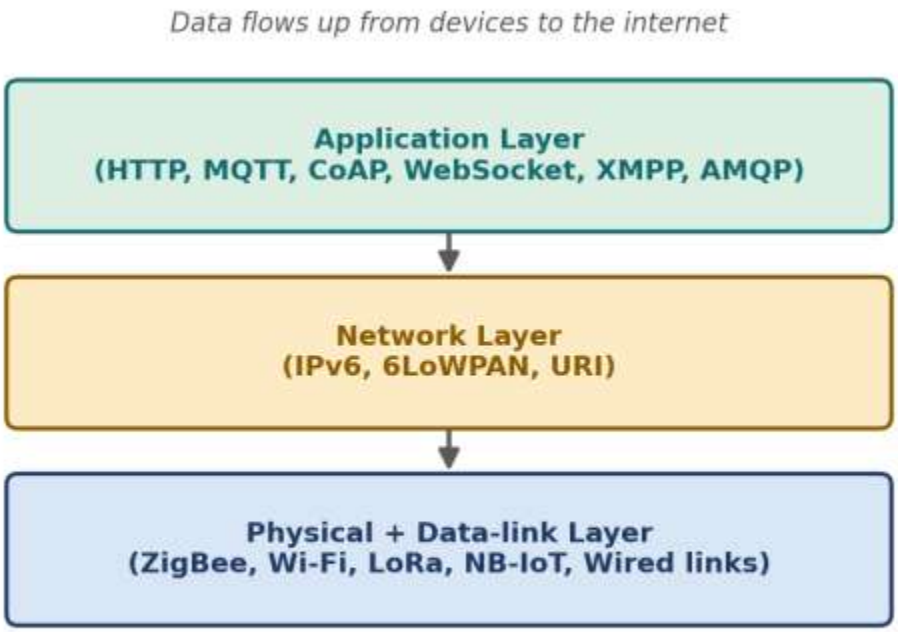


Figure: How data moves up through the physical/data-link, network, and application layers.

ANALOGY

Think of a school. Each classroom (local network) has its own way for students to pass notes — some rooms whisper (short-range, low-power like ZigBee), some use a megaphone (long-range like LoRa). The class monitor (gateway) then walks the collected notes to the principal's office (the internet).

EXAMPLE

A smart home has a temperature sensor, a smart bulb, and a door lock — each talks over Wi-Fi or ZigBee to a central hub, which then sends the combined data to the cloud over the home's internet connection.

2. ZigBee (IEEE 802.15.4)

ZigBee is a short-range, low-power wireless technology built on the IEEE 802.15.4 standard. It is designed for devices that send small amounts of data occasionally and must run for months or years on a small battery — think sensors, remote controls, and smart-home actuators. ZigBee devices form a WPAN (Wireless Personal Area Network).

Key Points

- Maximum frame size at Layer-1 (physical/MAC) = 127 Bytes — ZigBee is meant for tiny messages, not big files.
- A device can work in six roles: end point, ZigBee router, network coordinator, ZigBee-IP coordinator, ZigBee-IP router, and IP host.
- ZigBee IP adds IPv6 support so ZigBee devices can connect directly to internet-based systems.
- Network is self-configuring and self-healing (a mesh network) — if one path breaks, data automatically finds another route.
- Range: 10–200 m; Data rate: 250 kbps; Security: AES-CCM-128 encryption; Latency: less than 10 ms.
- Topologies supported: Star, Mesh, and combinations forming bigger networks via multiple star topologies.

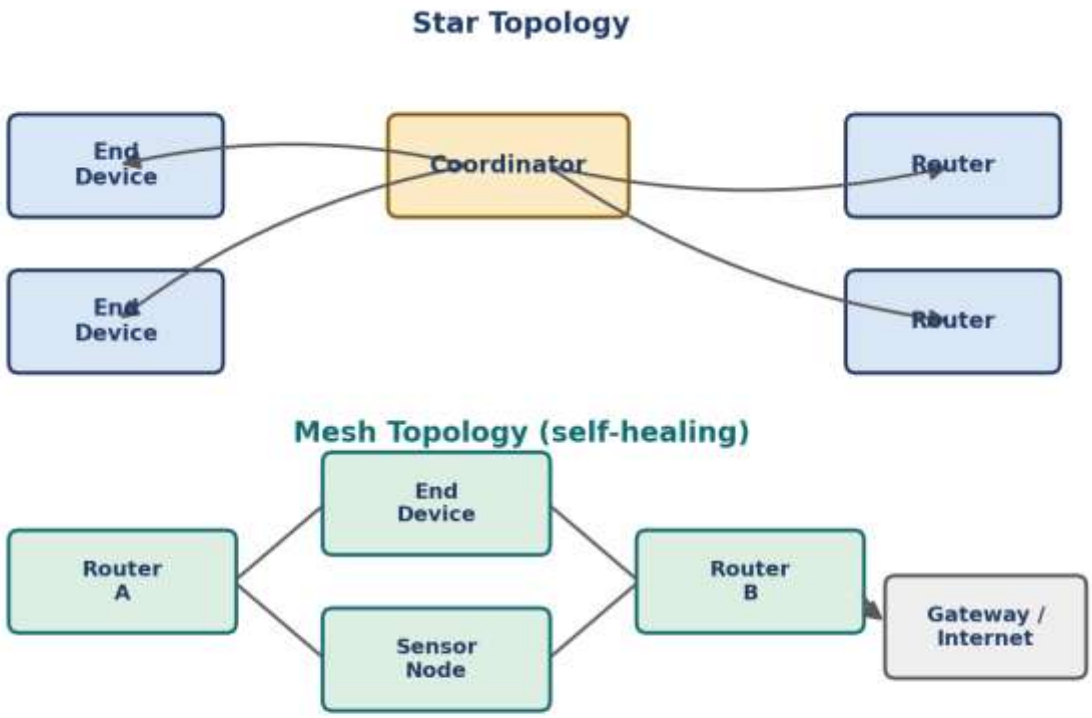


Figure: ZigBee star topology (left) vs. self-healing mesh topology (right).

ANALOGY

ZigBee is like a bucket-brigade at a fire — instead of one person running the water bucket the whole distance (long-range, high power), each person (device) only needs to pass it to the next person nearby. This 'hop by hop' passing (mesh routing) is efficient and lets the network heal itself if one person steps out.

EXAMPLE

Smart light bulbs in a house form a ZigBee mesh: turning off the hallway light still lets the living-room light relay commands to the bedroom light, even if the hub's direct wireless signal doesn't reach that far.

3. Wi-Fi (IEEE 802.11)

Wi-Fi is the most familiar wireless technology — it creates a Wireless Local Area Network (WLAN) using the IEEE 802.11 family of standards. It connects everything from laptops to smart TVs and IoT sensors to a router or Access Point (AP), which then bridges the connection to the wired internet.

Key Points

- Devices connect via a Base Station/Access Point, or directly to each other in peer-to-peer mode (no AP) called an IBSS (Independent Basic Service Set).
- A group of devices connected via one AP is called a BSS (Basic Service Set), identified by an SSID service set identifier (the Wi-Fi network name you see on your phone).
- Common variants: 802.11b (11 Mbps, 30 m), 802.11g (54 Mbps), 802.11n (up to 600 Mbps, uses multiple antennas — MIMO).
- Operates on 2.4 GHz or 5 GHz frequency bands; coverage range is roughly 30–125 m.
- Security is handled using WPA (Wi-Fi Protected Access) and the older, weaker WEP (Wired Equivalent Privacy).
- Downsides: higher power consumption than ZigBee/6LoWPAN, and prone to interference from walls and other 2.4 GHz devices.

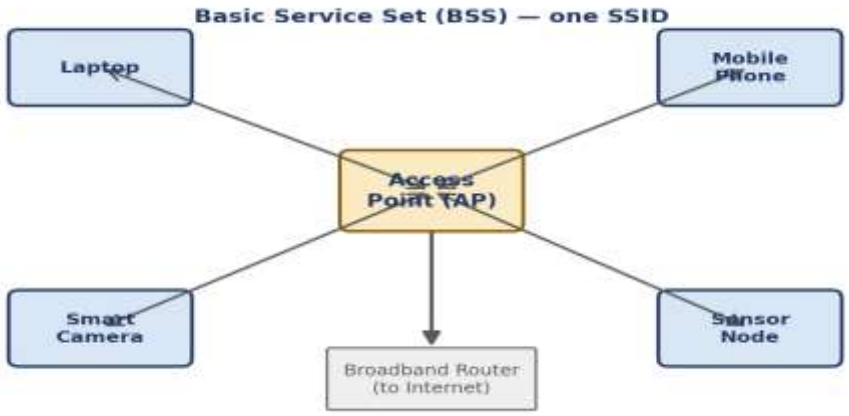


Figure: A Basic Service Set (BSS) — devices connect to one Access Point, which bridges to the internet.

ANALOGY

Wi-Fi is like a large classroom where the teacher (Access Point) stands in the middle. Every student (device) can hear the teacher and speak to the teacher directly, but students far from the teacher (weak signal near walls) start struggling to hear clearly.

EXAMPLE

A smart CCTV camera streams high-resolution video continuously over Wi-Fi because it needs high bandwidth (54+ Mbps) — something a low-power protocol like ZigBee simply cannot support.

4. 6LoWPAN — IPv6 over Low-Power Networks

6LoWPAN **(IPv6 over Low-power Wireless Personal Area Networks)** is an adaptation layer that sits between the IEEE 802.15.4 physical layer and the IPv6 network layer. Its job is simple but crucial: IPv6 packets are large (up to 1280 Bytes), but 802.15.4 frames are tiny (127 Bytes max) — 6LoWPAN compresses and fragments IPv6 packets so they can travel over these low-power radios, then reassembles them at the destination.

Key Points

- Core features: header compression, fragmentation, and reassembly.
- First fragment header = 27 bits (11 bits datagram size + 16 bits datagram tag); later fragments' header = adds an 8-bit offset field.
- Fragment reassembly time-out is typically set to 60 seconds — if all pieces don't arrive in time, the packet is dropped.
- Frame size math: IPv6 header (40 B) + UDP header (8 B) + MAC address (25 B) + AES-128 security (21 B) = 94 B overhead, leaving only about 33 Bytes for actual application data out of the 127 B frame.
- Also standardizes IETF-recommended neighbour discovery (6LoWPAN-ND) and supports mesh routing (via RPL — Routing Protocol for Low-power and lossy networks).

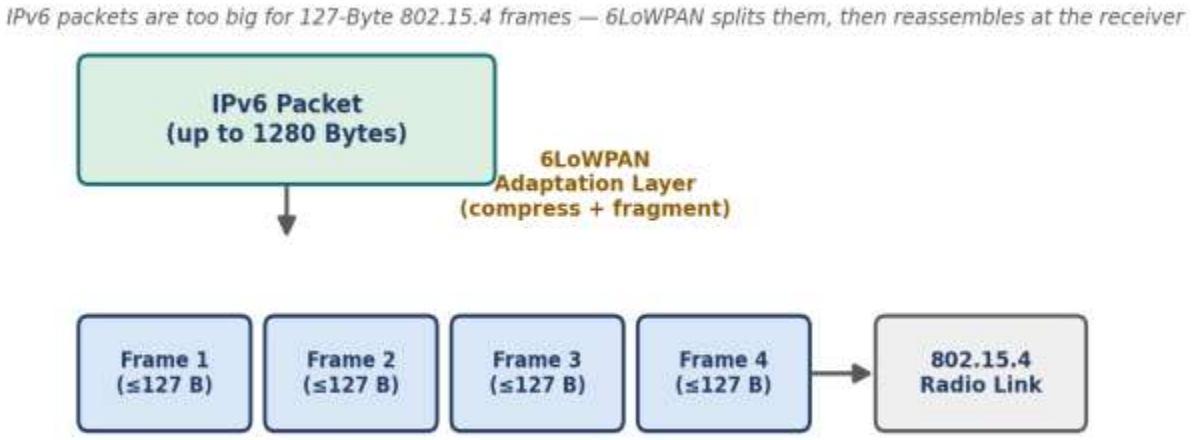


Figure: A large IPv6 packet is fragmented into 127-Byte frames for transmission, then reassembled.

ANALOGY

Imagine mailing a large parcel (a 1280-Byte IPv6 packet) but the delivery van can only carry small envelopes (127-Byte 802.15.4 frames). 6LoWPAN is the courier company that chops the parcel into many small envelopes, labels each one with its position (fragmentation), and reassembles them into the original parcel once they all arrive.

EXAMPLE

A soil-moisture sensor in a farm sends an IPv6-based report to the cloud. Because the radio link can only carry 127 B at a time, 6LoWPAN splits the report into small fragments, sends them one by one, and the receiving gateway reassembles the full IPv6 packet before forwarding it onward.

5. LPWAN Technologies: LoRa and NB-IoT

LPWAN (Low-Power Wide Area Network) technologies are designed for devices that need to send small amounts of data over very long distances (kilometers) while running for years on a battery. LoRa and NB-IoT are the two leading LPWAN choices, but they take different approaches.

Key Points

- LoRa uses unlicensed spectrum — anyone can deploy a LoRa network without paying a telecom operator, making it open, cheap, and flexible.
- NB-IoT (Narrowband IoT) uses licensed cellular spectrum (run by telecom operators), giving it lower latency, better security, and guaranteed quality — but at higher cost.
- LoRa best fits: smart agriculture, smart cities, asset tracking — cost-sensitive, less security-critical.
- NB-IoT best fits: smart metering, supply-chain tracking, healthcare monitoring — needs high reliability and security.

ANALOGY

LoRa is like using a free public walkie-talkie channel — anyone can set it up cheaply, but there's no guarantee of exclusive access or top-tier security. NB-IoT is like renting a private, secured phone line from a telecom company — it costs more, but you get guaranteed quality and protection.

EXAMPLE

A city-wide network of parking sensors uses LoRa (cheap, no license fees needed for thousands of sensors), while a hospital's remote patient-monitoring device uses NB-IoT because patient data needs guaranteed delivery and strong security.

II. Overview of IoT Application-Layer Protocols

Once data reaches the internet (network layer), it needs an application-layer protocol to actually structure and deliver the message to the right software application. IoT commonly uses six such protocols: HTTP, MQTT, CoAP, WebSocket, XMPP, and AMQP — each optimized for different situations (constrained device vs. web server, one-time request vs. constant streaming, etc.).

Key Points

- HTTP/HTTPS — the standard web protocol; simple but relatively heavy for tiny IoT devices.
- MQTT — lightweight publish/subscribe protocol, ideal for constrained IoT devices.
- CoAP — a 'lightweight HTTP' designed specifically for constrained networks and devices.
- WebSocket — enables real-time, bidirectional communication over a single connection.
- XMPP — XML-based protocol originally for chat/instant messaging, now used in some IoT contexts.
- AMQP — enterprise-grade message-queuing protocol focused on reliability and interoperability.

ANALOGY

Think of these as different types of postal services: HTTP is like registered post (reliable but you fill a new form for every letter), MQTT is like a newsletter subscription (publisher sends once, all subscribers automatically receive it), and WebSocket is like an open phone line where both sides can talk anytime without hanging up and redialing.

EXAMPLE

A weather-monitoring IoT project might use MQTT for the sensors (low power, frequent tiny updates) and HTTP for the dashboard website that a user opens in a browser to view the data.

a. MQTT — Message Queuing Telemetry Transport

MQTT is a lightweight, open, publish/subscribe messaging protocol built specifically for M2M and IoT communication, where devices may have limited bandwidth or unreliable connections. 'Telemetry' means measuring something and sending the value to a remote location — exactly what IoT sensors do. The current common version is MQTT v3.1.1, standardized by OASIS in 2014. A slimmed-down version, MQTT-SN (Sensor Networks), extends MQTT to non-TCP/IP networks like ZigBee.

Key Points

- Follows a publish/subscribe (pub/sub) model, not the request/response model used by HTTP.
- Devices are Publishers (send data) or Subscribers (receive data), connected through a central MQTT Broker.
- The Broker stores-and-forwards messages, matches subscriptions to publications, and routes messages to the correct subscriber.
- Supports 'retain' — the broker keeps the last message on a topic so a newly-connected subscriber immediately gets the latest value.
- Security: authentication via username/password at connect time, plus SSL/TLS encryption for the connection.
- Keeps track of client connection status until an explicit DISCONNECT message — enabling 'last will' type behaviour for dropped connections.

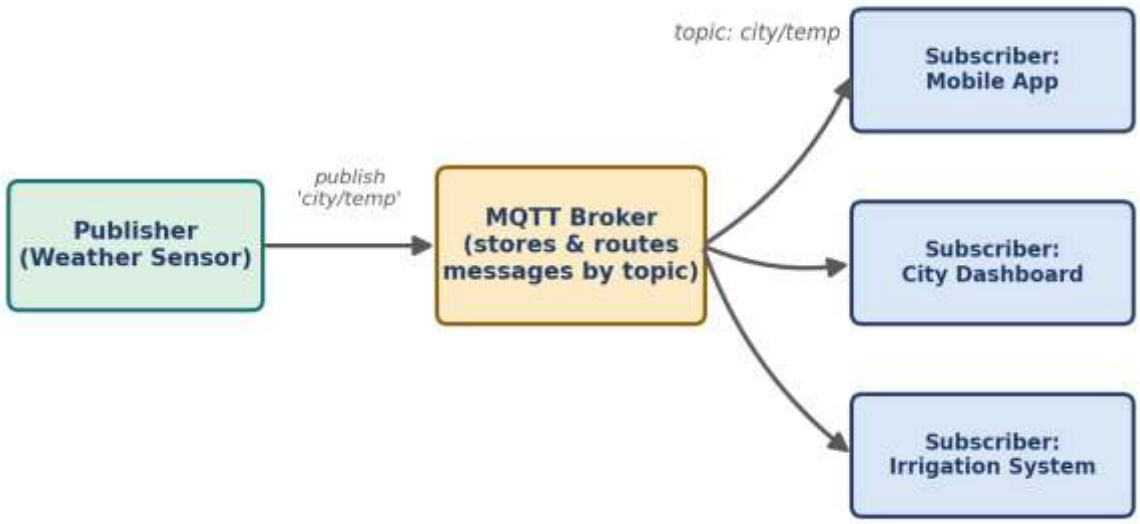


Figure: MQTT publish/subscribe model — the broker routes messages by topic to all interested subscribers.

ANALOGY

MQTT is exactly like subscribing to a YouTube channel. The creator (publisher) uploads a video (data) once. YouTube (the broker) doesn't send it directly to every viewer's phone individually asking 'do you want this?' — instead,

everyone who subscribed to that channel (topic) automatically receives it. You don't even need to know who else is subscribed.

EXAMPLE

A weather station publishes temperature readings to the topic 'city/temperature'. Any number of apps — a mobile weather app, a farmer's irrigation controller, a city dashboard — can subscribe to 'city/temperature' and instantly receive updates whenever new data is published, without the weather station needing to know who's listening.

b. WebSocket

WebSocket (defined in RFC 6455) provides full-duplex (two-way, simultaneous) communication over a single, long-lived TCP connection. Unlike HTTP, where the client must repeatedly ask the server 'anything new?' (polling), WebSocket keeps the connection open so either side can send data the instant it's ready.

Key Points

- Enables bidirectional communication over one TCP connection — no repeated handshakes needed.
- Very small header: just 2 Bytes (extendable to 4 Bytes) compared to HTTP's 100+ Byte headers — much lower latency.
- Handshake starts as an HTTP 'upgrade' request, so it works over existing web infrastructure (uses port 80 for ws:// and port 443 for wss:// which is TLS-secured).
- Removes the need for periodic HTTP polling, which reduces both latency and unnecessary network traffic.

ANALOGY

HTTP polling is like repeatedly calling a friend every 2 minutes asking 'any news yet?' — most calls waste time with 'no, nothing new.' WebSocket is like staying on a single open phone call — the moment your friend has news, they simply tell you, right then, without you having to ask.

EXAMPLE

A live IoT dashboard that shows real-time machine sensor readings uses WebSocket — the server pushes new readings the instant they arrive, giving a smooth live-updating graph instead of a page that refreshes every few seconds.

c. HTTP and HTTPS

HTTP (HyperText Transfer Protocol) is the standard protocol of the web — a client (browser or IoT device) requests a resource identified by a URL, and a server responds. HTTPS is the secure version, wrapping HTTP inside TLS/SSL encryption.

Key Points

- HTTP uses port 80 by default; HTTPS uses port 443.
- HTTP is stateless — every request is treated as brand new, with no memory of previous requests (no session data retained automatically).
- It behaves like a file-transfer protocol but is lighter and faster than protocols like FTP.
- It's flexible — if a connection breaks mid-transfer, the client can simply reconnect and retry.
- It's built around an Object-Oriented model — 'methods' (like GET, POST) act on 'objects' identified by a URL, similar to calling functions on objects in OOP.

ANALOGY

HTTP being 'stateless' is like a shopkeeper with total amnesia — every time you walk in, they treat you as a first-time customer, even if you were there five minutes ago. That's why websites need extra tools like cookies or sessions to 'remember' you across requests.

EXAMPLE

When your phone loads a web dashboard showing your smart thermostat's temperature, it sends an HTTPS GET request to a server, which responds with the current temperature data — a classic one request → one response cycle.

d. AMQP — Advanced Message Queuing Protocol

AMQP is an open-standard, application-layer protocol designed for enterprise-grade, message-oriented middleware. It focuses on reliable message queuing, routing, and interoperability between systems built by different vendors.

Key Points

- Defines how messages are exchanged between applications, with strong guarantees on delivery.
- Provides queuing, routing, and security features suited to business-critical messaging.
- Enables interoperability — a client built by one vendor can talk to a broker built by a completely different vendor, because they follow the same open standard.

ANALOGY

AMQP is like a formal, tracked courier service (compared to MQTT's casual newsletter-style delivery) — every package (message) is logged, routed carefully, and guaranteed to reach its destination, which matters a lot in banking or enterprise systems.

EXAMPLE

A bank's fraud-detection system uses AMQP to reliably pass transaction alerts between multiple internal software systems, where losing even a single message could be costly.

e. CoAP — Constrained Application Protocol

CoAP is a specialized, lightweight web-transfer protocol built specifically for resource-constrained IoT devices — those with very limited processing power, memory, and battery life. It mimics HTTP's request/response style but is optimized to run efficiently over low-power, low-bandwidth networks.

Key Points

- Uses a request/response model, similar in spirit to HTTP, but far lighter in overhead.
- Runs over UDP instead of TCP (unlike HTTP), reducing connection overhead for constrained devices.
- Asynchronous — it doesn't need to keep a connection open and waiting the way some other protocols do.

ANALOGY

If HTTP is a formal, detailed business letter, CoAP is the same message written as a short, efficient text — same intent (request → response), but stripped of all unnecessary formality so it costs less 'battery and bandwidth' to send.

EXAMPLE

A battery-powered soil sensor in a remote field uses CoAP to respond to a 'GET moisture-level' request from a farm-management server, since CoAP's low overhead helps the sensor's battery last much longer than if it used full HTTP.

f. XMPP — Extensible Messaging and Presence Protocol

XMPP is an open, XML-based protocol originally developed for the Jabber instant-messaging project. It supports near-real-time exchange of structured data and has since expanded beyond chat into voice/video calls and IoT messaging.

Key Points

- Based on XML, enabling structured, extensible data exchange between network entities.
- Originally built for instant messaging, presence information (online/offline status), and contact-list management.
- Now adapted for various uses including chat, calls, and select IoT messaging scenarios.

ANALOGY

XMPP works much like a group chat app — it not only delivers messages but also tracks 'presence' (who's online, offline, or busy), which is useful for IoT devices too (is this sensor currently online and reachable?).

EXAMPLE

An IoT platform might use XMPP-style presence tracking to instantly know when a connected device goes offline (e.g., loses power or network), similar to how a chat app shows a contact as 'offline.'

3. Identification Protocols -- IPv6 and URI

For devices to be found and addressed on the internet, they need an identification system. IPv6 (Internet Protocol version 6) provides the addressing scheme, while a URI/URL scheme defines how a specific resource on a device or server is accessed.

Key Points

- IPv6 provisions a vastly larger address space than IPv4 — essential since billions of IoT devices each need a unique address.
- Supports hierarchical address allocation, enabling efficient route aggregation and smaller routing tables across the internet.
- Manages device mobility, security, and auto-configuration more natively than IPv4.
- Supports multicast addressing (sending to a group efficiently) and 'jumbograms' (very large datagrams).
- A URI (Uniform Resource Identifier)/URL scheme (like http:// or https://) specifies exactly how a browser or client should access a resource, and HTTPS adds encryption on top of HTTP.

ANALOGY

If IPv6 addresses are like giving every single grain of sand on Earth its own unique postal address (that's how big the address space is), then a URI is like the specific 'apartment number and door' instructions for finding one exact resource once you've reached the right address.

EXAMPLE

A smart streetlight might have a unique IPv6 address so the city's control system can reach it directly, while the specific URL https://city.gov/streetlight42/status specifies exactly which resource (its status page) to fetch.

4. Network Architecture: WSN and Cellular (4G/5G)

Beyond individual protocols, IoT systems are built on top of broader network architectures — the overall blueprint of how devices connect, communicate, and share data. Two major building blocks are Wireless Sensor Networks (WSN) for local sensing, and cellular networks (4G/5G) for wide-area, high-speed connectivity.

Key Points

- WSN (Wireless Sensor Network): a self-organizing network of spatially distributed sensor nodes that cooperatively monitor physical/environmental conditions and send data to a central location (e.g. a gateway).
- WSN key characteristics: decentralized & self-organizing, distributed sensing, wireless communication.
- WSN components: Sensor Nodes (the many small devices that sense data) and a Gateway (which collects and forwards data onward).
- 4G: faster data speeds (up to ~100 Mbps), low latency, an all-IP network, and technologies like LTE and MIMO for better performance; enables VoLTE (Voice over LTE).
- 5G: the next generation after 4G — even faster data speeds, even lower latency, and much greater device connectivity density — well suited for massive IoT deployments.

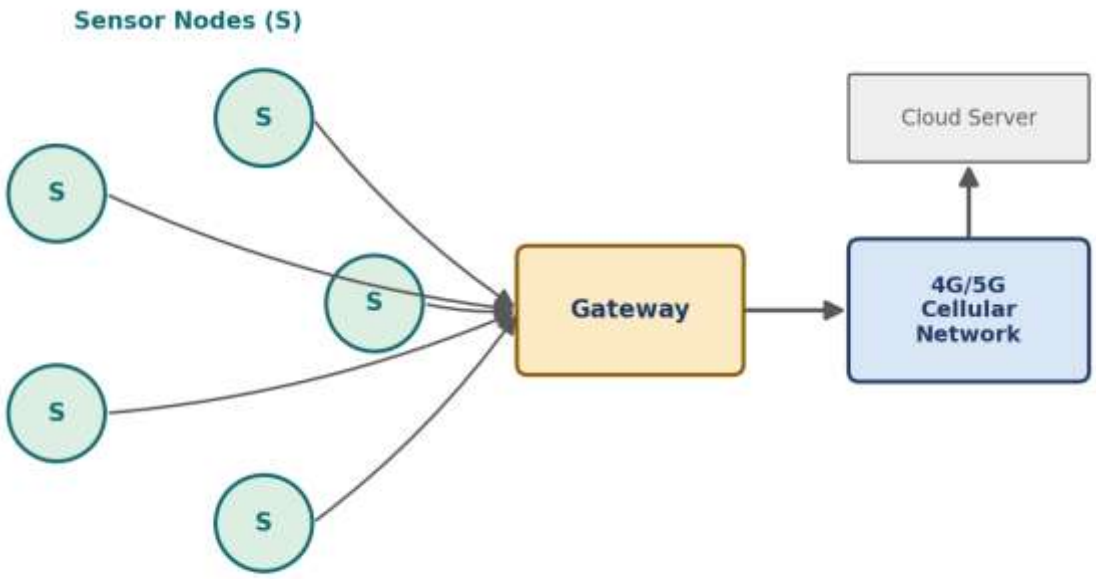


Figure: Sensor nodes forward data to a gateway, which uses 4G/5G to reach the cloud.

ANALOGY

A WSN is like a class of students spread across a large field, each shouting out small pieces of information about their surroundings (temperature, motion) to one class monitor (gateway) standing nearby, who compiles everything into one report. 4G/5G, by contrast, is like a super-highway that carries that compiled report at high speed all the way to the head office (cloud), no matter how far away it is.

EXAMPLE

In a smart-agriculture project, a WSN of soil and humidity sensors scattered across a farm gathers local data and forwards it to a gateway, which then uses a 4G/5G cellular connection to send the aggregated data to a cloud server for analysis — combining short-range sensing with long-range connectivity.